📊 Retail Transaction Data Analysis & KPI Dashboard Development Using SQL

📌 Project Description

This project presents a comprehensive end-to-end analysis of retail transaction data to evaluate sales performance, customer behavior, and business efficiency. The primary objective was to transform raw transactional data into structured insights that support data-driven decision-making.

The project integrates SQL for data cleaning and analysis, Excel for KPI dashboard development, Miro for workflow planning, and Canva for professional presentation.

🎯 Objectives

The key objectives of the project were:

- To calculate total revenue and key business KPIs
- To analyze sales performance across locations
- To identify high-value customer categories
- To determine best-selling products
- To evaluate payment method usage patterns
- To assess the impact of discounts on spending behavior

🛠 Tools & Technologies Used

SQL (MySQL Workbench)

- Data cleaning and validation
- Aggregation queries
- KPI calculations
- Revenue and performance analysis
- Ranking and customer lifetime value analysis

Microsoft Excel (Advanced)

- Pivot Tables
- Interactive KPI Dashboard
- Data Visualization
- Slicers & Filters
- Time-Series Trend Analysis
- Conditional Formatting

Miro

- Project workflow planning
- KPI structuring
- Problem definition mapping

Canva

- Executive-level business presentation
- Insight communication and storytelling

🔄 Project Workflow

- Imported raw CSV data into SQL database.
- Performed data cleaning, validation, and correction of missing values.
- Developed SQL queries to compute revenue, customer segmentation, product demand, and discount impact.
- Exported structured results into Excel.
- Designed an interactive dashboard using pivot tables and KPI metrics.
- Generated business insights and strategic recommendations.
- Presented findings using a structured slide presentation.

📊 Key Analysis Performed

- Total Revenue & Average Transaction Value
- Revenue by Location
- Revenue by Customer Category
- Top 10 Best-Selling Products
- Payment Method Distribution
- Monthly Revenue Trend
- Discount vs Non-Discount Spending Comparison
- Customer Lifetime Value (Advanced Analysis)

💡 Key Insights

- Identified top-performing locations contributing the majority of revenue.
- Determined high-value customer segments for targeted marketing.
- Recognized best-selling products for optimized inventory management.
- Evaluated whether discount strategies increase transaction value.
- Highlighted underperforming areas requiring operational improvement.

📈 Business Impact

- The analysis provides actionable insights that can support:
- Revenue growth strategies
- Inventory optimization
- Marketing targeting
- Discount policy evaluation
- Operational performance improvement

🚀 Project Outcome

This project demonstrates the ability to manage the complete analytics lifecycle — from raw data ingestion to insight-driven business recommendations. It reflects practical skills in SQL querying, data transformation, KPI development, dashboard design, and business intelligence reporting.